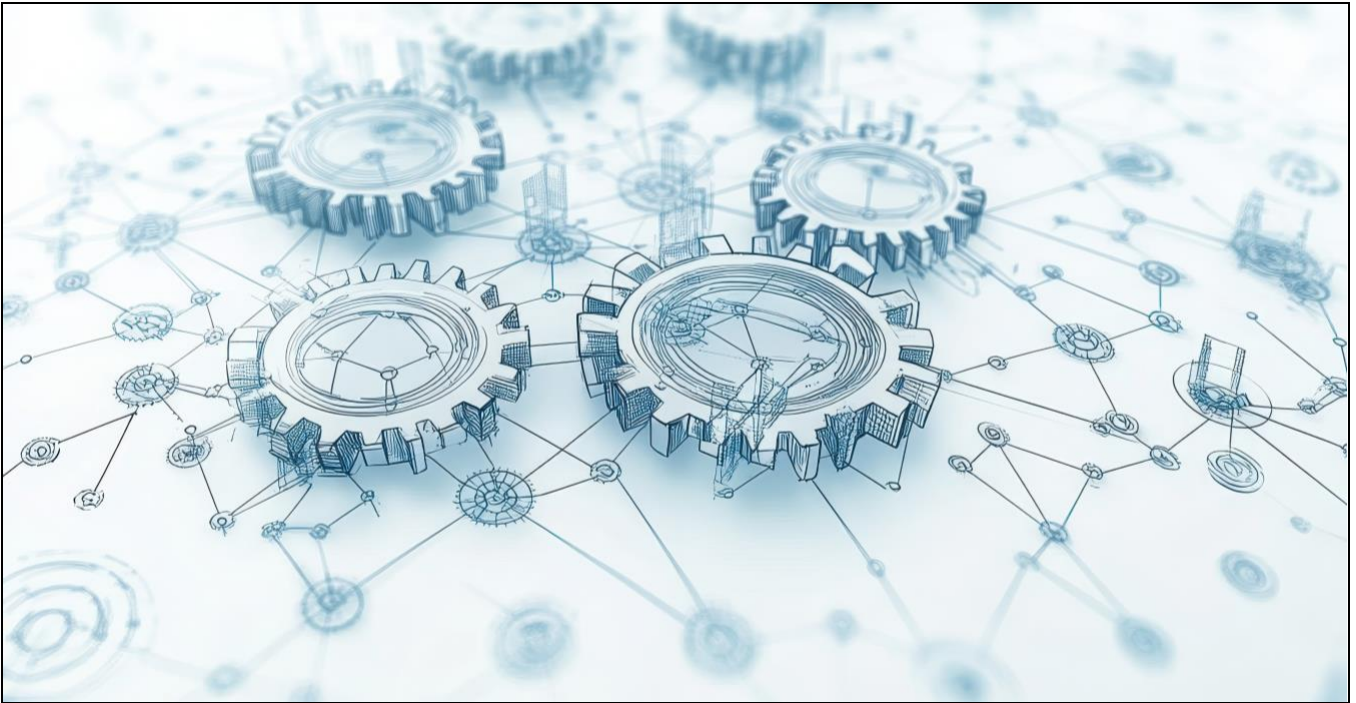


Level 4

Watsonx Orchestrate

Hands-on lab guide: Agent knowledge



Version 3.0

February 2026

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Introduction

Lab overview

In watsonx Orchestrate, agents can use knowledge bases to search for relevant, grounded content or answers.

With the watsonx Orchestrate Agent Development Kit (ADK 2.4), you can define knowledge bases using YAML or JSON, and import them to your active environment using the ADK 2.4 CLI.

You can create two types of knowledge bases: built-in and external. Built-in knowledge bases are created using a built-in Milvus instance and are populated with documents you import. External knowledge bases refer to your own Milvus or Elasticsearch instances, which you can connect to watsonx Orchestrate for use by agents.

This lab will guide you through the steps to use the built-in Milvus instance to create your knowledge base. When complete, the agent will be able to respond any questions related to the content of the pdf documents. In this lab, we will use two pdf documents:

- The history of IBM
- The exported IBM Wikipedia page (Source: <https://en.wikipedia.org/wiki/IBM>)

The agent will provide citations describing the source of information the response is based on and the precise location in the source document as shown below:

The screenshot displays the IBM watsonx Orchestrate web interface. On the left sidebar, the 'Agents' section shows 'ibm_agent' selected. Below it, the 'Chats' section shows a list of chat titles, with 'What was the origin...' highlighted. The main chat area shows a conversation with 'ibm_agent'. The user's question is 'What was the original creation name of IBM?'. The agent's response is 'The original creation name of IBM was the Computing-Tabulating-Recording Company (CTR)'. Below the response, a citation box is shown for 'IBM_wikipedia.pdf', containing a snippet of text from a Wikipedia page dated June 16, 1911, describing the formation of the CTR. A 'View source' link is provided for the citation. The interface also shows a 'Show Reasoning' dropdown and a '1/1' indicator for the citation. At the bottom, there is a text input field with the placeholder 'Type something...' and a send button.

What you will learn

Once you have completed this lab you will have developed skills enabling you to complete the following tasks:

1. Create a knowledge base in watsonx Orchestrate.
2. Create an agent that uses this knowledge base.
3. Use the agent description and agent instructions to define the agent's behavior.

Environment

This lab has been tested with the Agent Development Kit (ADK) version 2.4

How to get support

If you encounter an issue with this lab,

- IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.
- Business Partners can request help at the [Partner Plus Support](#) website.

Prerequisites

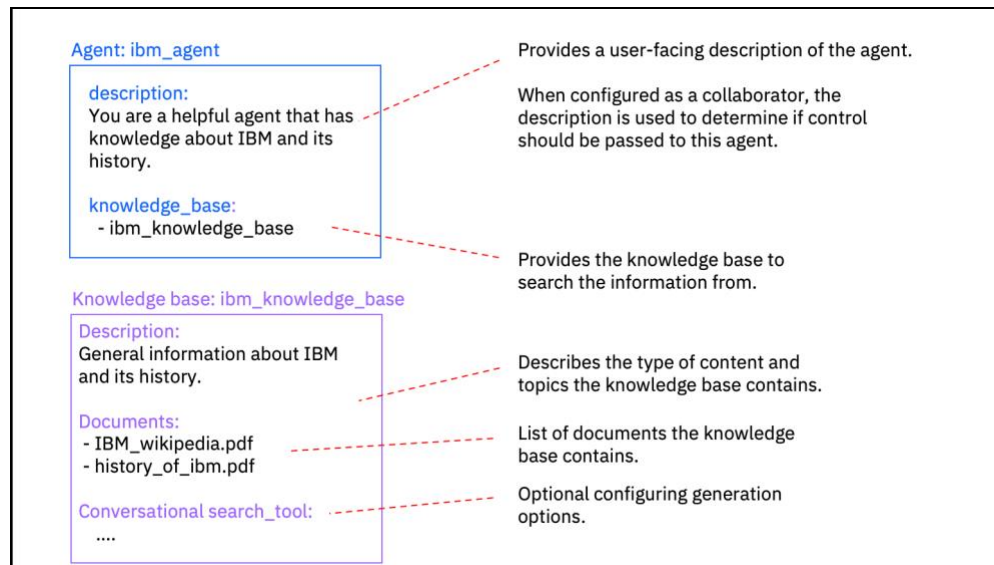
Make sure the following prerequisites are satisfied before proceeding with this lab:

- You have performed the Preparation Lab guide including the environment validation step
- You have your local ADK 2.4 environment up and running (Orchestrate server started after validating the environment in the preparation guide OR after having performed another ADK related lab)

Note: If for some reason your server is not started correctly, reset your server following the [Reset your environment](#) appendix.

IBM knowledge base agent

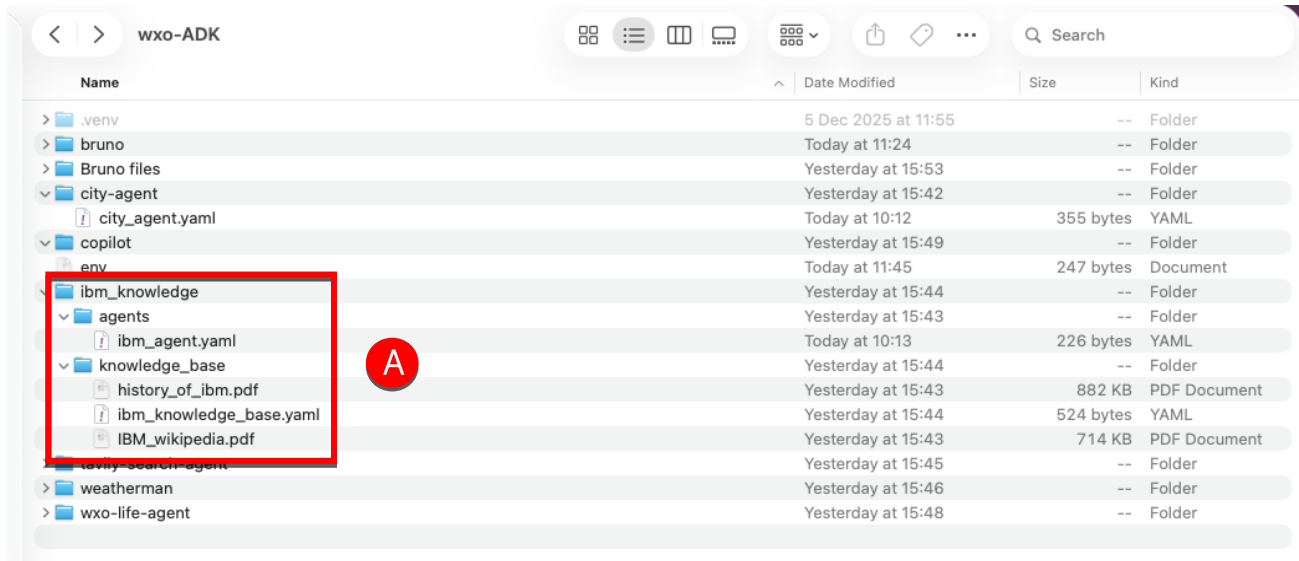
In this section you will create and test an IBM agent. You will create the agent and provide this agent with a knowledge base populated with two documents. The diagram below provides an overview of the configuration used in this section. Review the agent and knowledge base configuration shown below before proceeding.



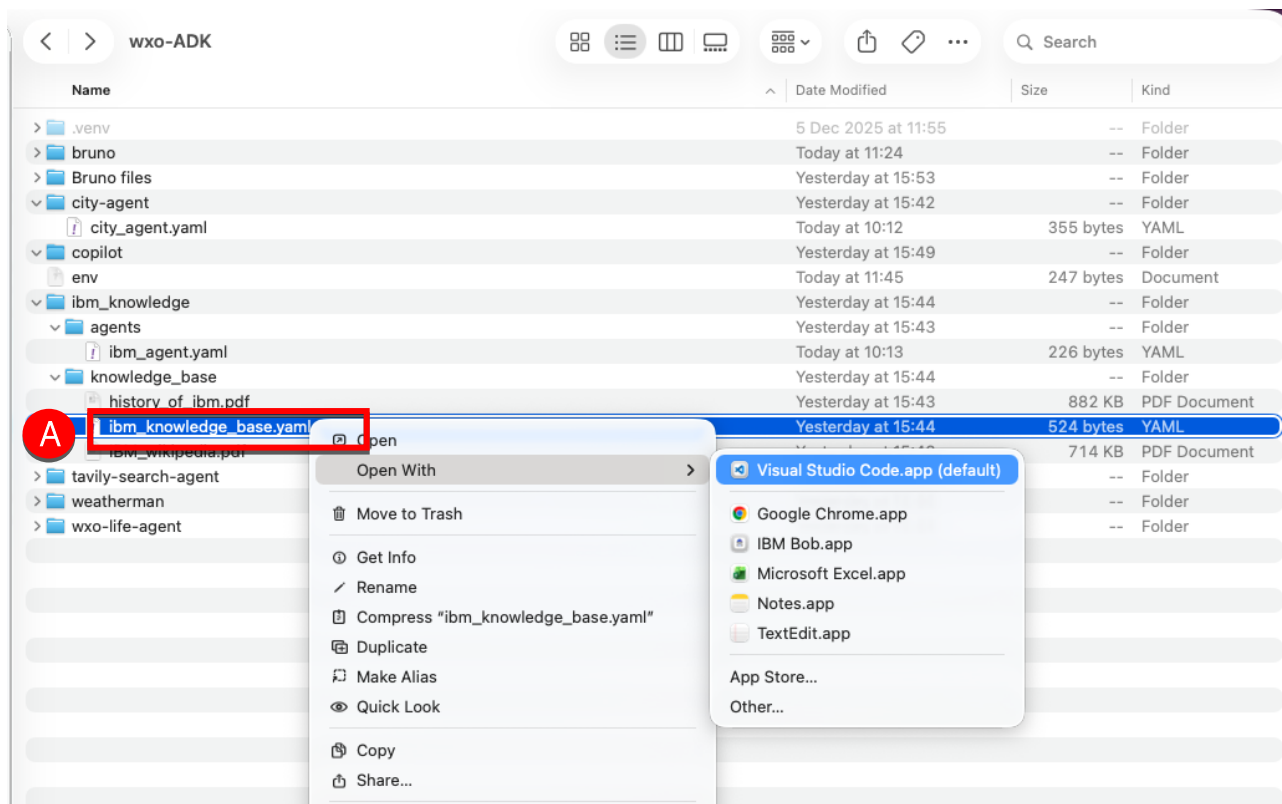
Knowledge base creation

In this section you will learn how to use the watsonx Orchestrate ADK to create a knowledge base in your watsonx Orchestrate instance. This knowledge base will then be used by an agent to answer topic specific questions.

1. Open the ADK environment and start the watsonx Orchestrate server if not yet done. Refer to the [Reset your environment appendix](#) if your orchestrate server is not started.
2. Navigate in your `wxo-ADK/ibm_knowledge/Knowledge_base` folder (A).



3. Edit the `ibm_knowledge_base.yaml` file using your preferred text editor (A).



4. Review the implementation of the Knowledge base (this is explained below).

```

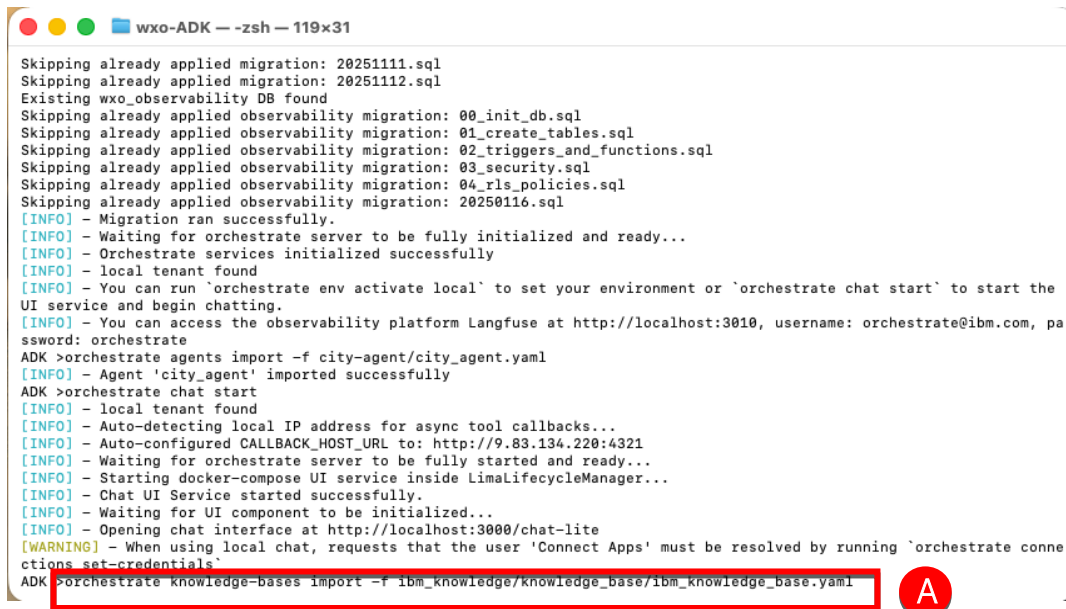
1 spec_version: v1
2 kind: knowledge_base
3 name: ibm_knowledge_base
4
5 description: General information about IBM and its history
6 documents:
7   - IBM_wikipedia.pdf
8   - history_of_ibm.pdf
9
10 conversational_search_tool:
11   generation:
12     prompt_instruction: Custom instruction
13     generated_response_length: Moderate
14   confidence_thresholds:
15     retrieval_confidence_threshold: Low
16     response_confidence_threshold: Low
17   query_rewrite:
18     enabled: True
19   citations:
20     citations_shown: -1
  
```

prompt_instruction	If specified, this instruction will be included in the prompt sent to the language model to guide response generation.
generated_response_length	Set this to one of Concise, Moderate or Verbose. This setting adjusts the prompt to request responses of the specified length. If not set, the default is Moderate.
retrieval_confidence_threshold	Set this to one of Lowest, Low, High or Highest. This threshold determines the minimum confidence required that the retrieved documents answer the user's query. If the confidence is below the threshold, the agent will return a default "I don't know" response instead of generating a response. The default is "Low".
response_confidence_threshold	Set this to one of Lowest, Low, High or Highest. This threshold evaluates the confidence that both the generated response and the retrieved documents answer the user's query. If the confidence is below the threshold, the agent will return a default "I don't know" response. The default is Low.
query_rewrite	If enabled, the user's query is rewritten using the context of the conversation to support multi-turn interactions. This setting is enabled by default.
citations_shown	This controls the maximum number of citations shown to the user in a knowledge-based response. If not set, the default is -1, which means all available citations will be displayed

Note: You can change the `prompt_instruction` parameter to add your own instructions for the content generation engine. We will use the default in this lab.

5. Enter the following command into the terminal (A) to create the knowledge base.

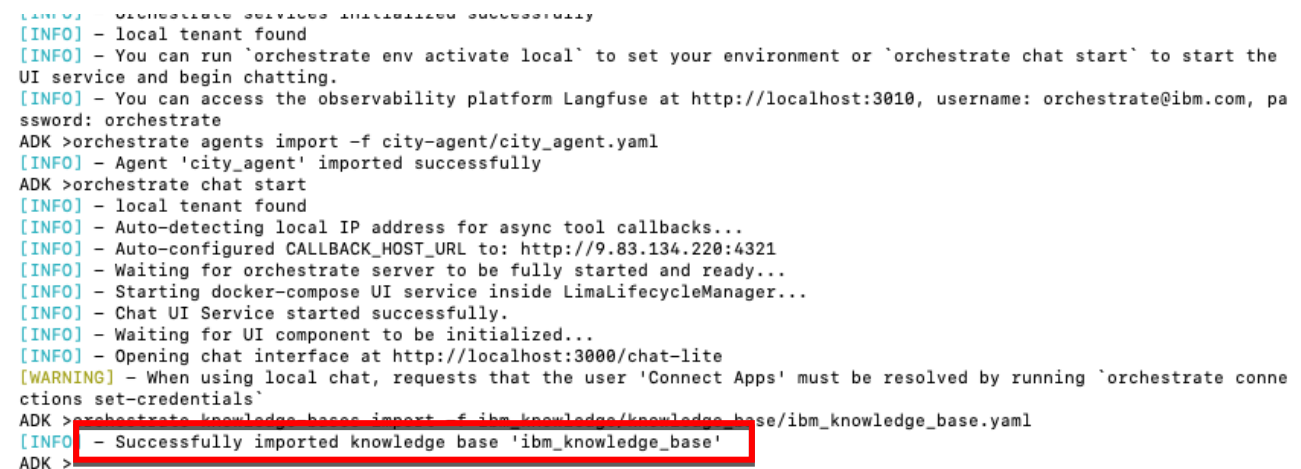
```
orchestrate knowledge-bases import -f
ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
```

A terminal window titled 'wso-ADK -- zsh -- 119x31' showing the output of the 'orchestrate' command. The output includes several status messages about migrations and service initialization. The command 'orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml' is entered and highlighted with a red box. A red circle with the letter 'A' is next to the terminal window.

```
wso-ADK -- zsh -- 119x31
Skipping already applied migration: 20251111.sql
Skipping already applied migration: 20251112.sql
Existing wso_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK >orchestrate agents import -f city-agent/city_agent.yaml
[INFO] - Agent 'city_agent' imported successfully
ADK >orchestrate chat start
[INFO] - local tenant found
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Waiting for orchestrate server to be fully started and ready...
[INFO] - Starting docker-compose UI service inside LimaLifecycleManager...
[INFO] - Chat UI Service started successfully.
[INFO] - Waiting for UI component to be initialized...
[INFO] - Opening chat interface at http://localhost:3000/chat-lite
[WARNING] - When using local chat, requests that the user 'Connect Apps' must be resolved by running 'orchestrate conne
ctions set-credentials'
ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
```

Note: On Windows, if you use folder name completion using the TAB key, the slashes are changed into back-slashes. The orchestrate command still finds the correct yaml file but then fails to find the PDF documents. Ensure you use forward slash “/”.

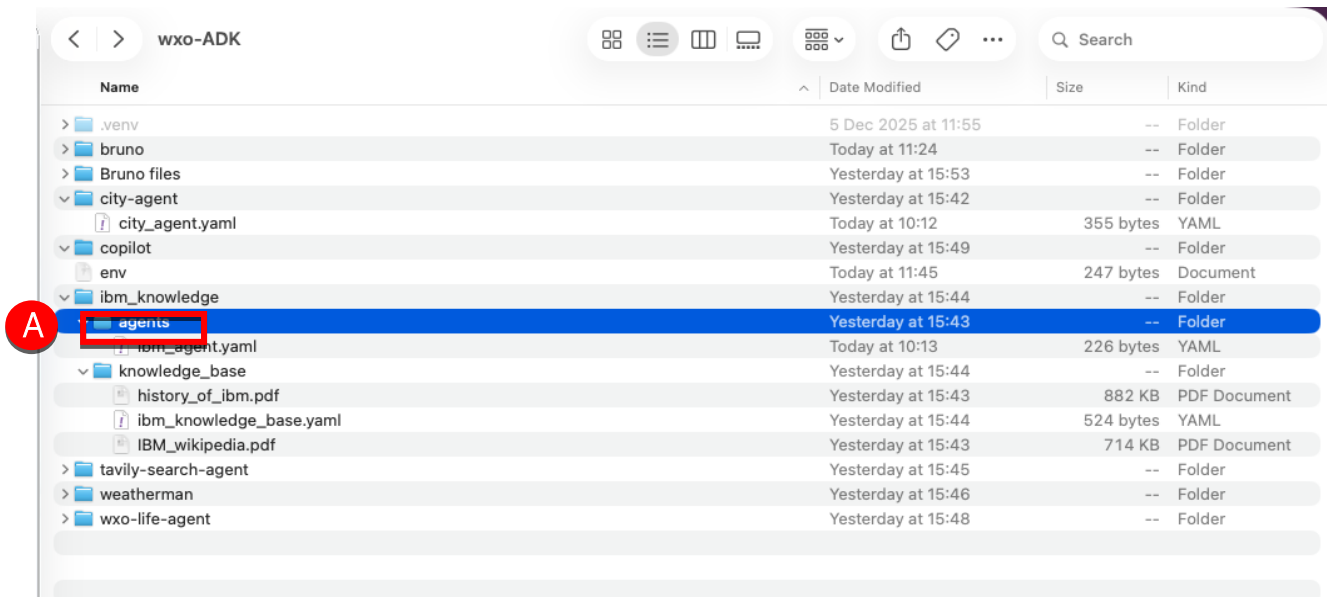
You should see the following message when the knowledge base is created:

A terminal window showing the output of the 'orchestrate' command. The command 'orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml' is entered and highlighted with a red box. The output shows a successful import message. A red circle with the letter 'A' is next to the terminal window.

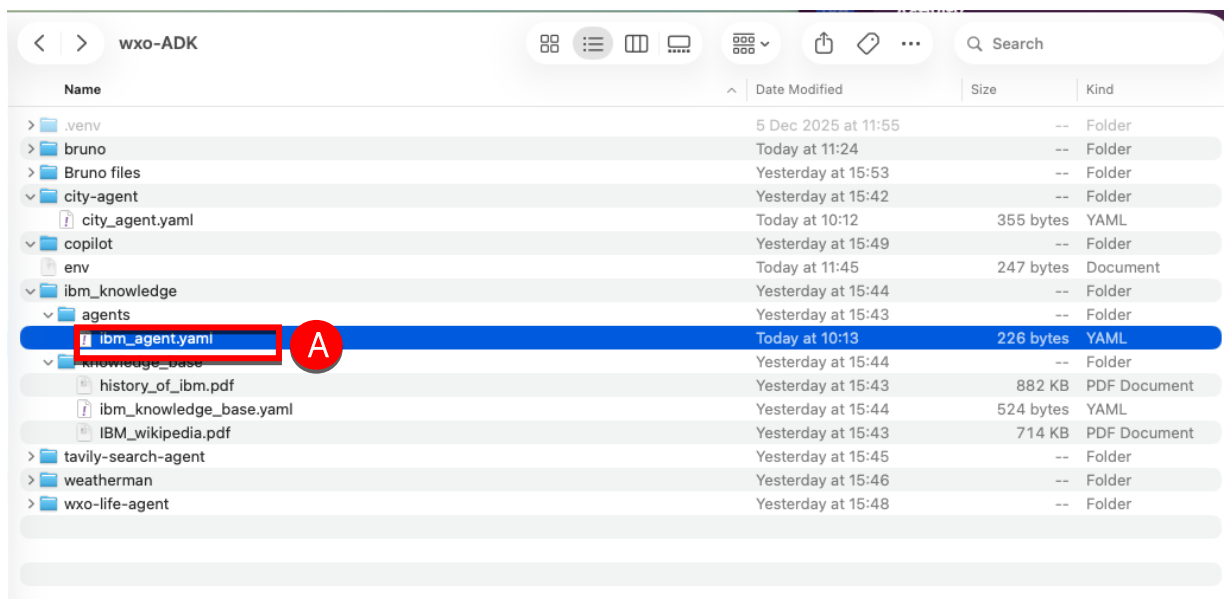
```
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK >orchestrate agents import -f city-agent/city_agent.yaml
[INFO] - Agent 'city_agent' imported successfully
ADK >orchestrate chat start
[INFO] - local tenant found
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Waiting for orchestrate server to be fully started and ready...
[INFO] - Starting docker-compose UI service inside LimaLifecycleManager...
[INFO] - Chat UI Service started successfully.
[INFO] - Waiting for UI component to be initialized...
[INFO] - Opening chat interface at http://localhost:3000/chat-lite
[WARNING] - When using local chat, requests that the user 'Connect Apps' must be resolved by running 'orchestrate conne
ctions set-credentials'
ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
[INFO] - Successfully imported knowledge base 'ibm_knowledge_base'
ADK >
```

Agent creation

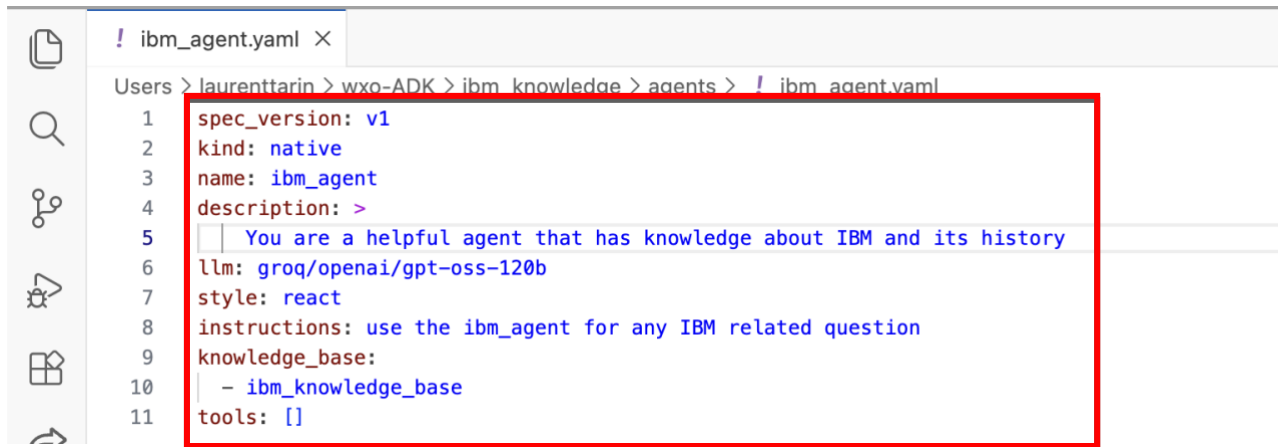
1. Navigate to your `wxo-ADK/ibm_knowledge/agents` (A) folder.



2. Edit the `ibm_agent.yaml` file(A).



- Review the definition of the agent (A).

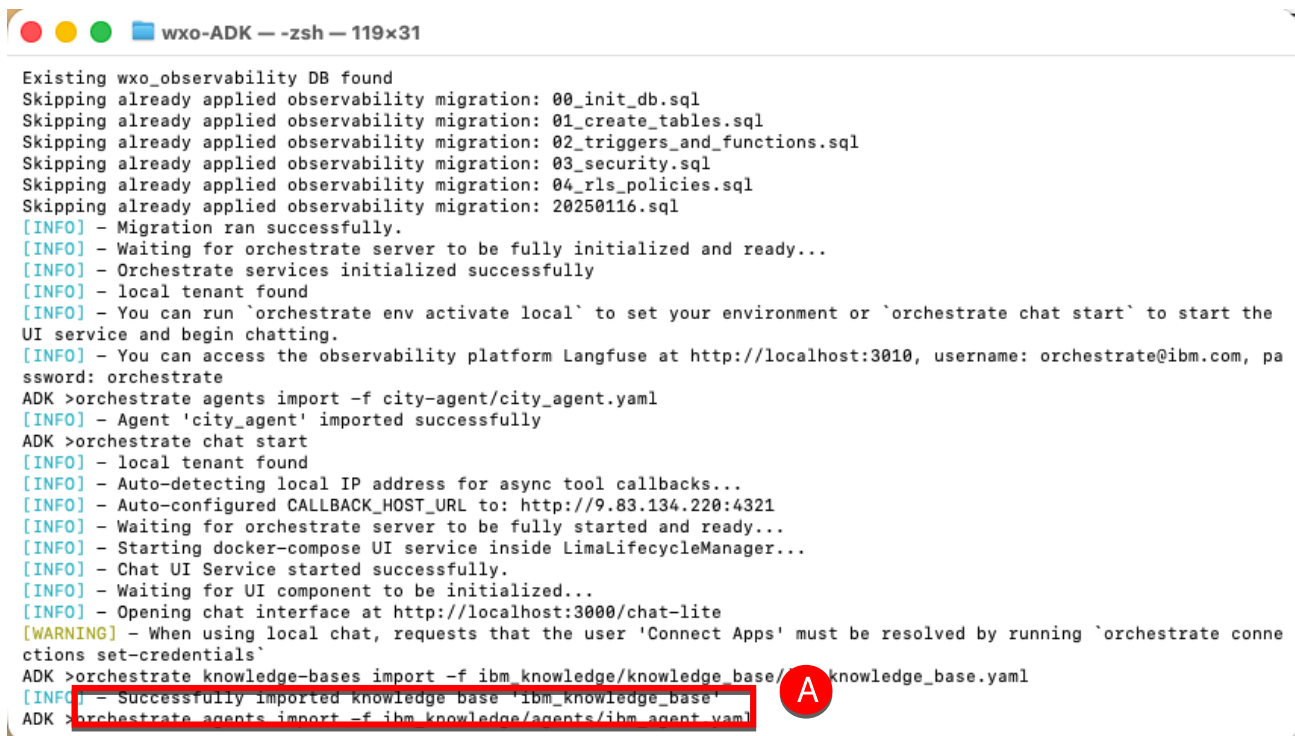


```
! ibm_agent.yaml X
Users > laurenttarin > wxo-ADK > ibm_knowledge > agents > ! ibm_agent.yaml
1 spec_version: v1
2 kind: native
3 name: ibm_agent
4 description: >
5   You are a helpful agent that has knowledge about IBM and its history
6 llm: groq/openai/gpt-oss-120b
7 style: react
8 instructions: use the ibm_agent for any IBM related question
9 knowledge_base:
10   - ibm_knowledge_base
11 tools: []
```

This yaml file defines the agent, its description behavior/instructions, and the LLM it will use for planning and reasoning. This agent has no collaborators and no tools, but a single knowledge base is declared.

- Enter the following command into the terminal (A) to create the agent.

```
orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
```



```
wxo-ADK --zsh -- 119x31
Existing wxo_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run `orchestrate env activate local` to set your environment or `orchestrate chat start` to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK > orchestrate agents import -f city-agent/city_agent.yaml
[INFO] - Agent 'city_agent' imported successfully
ADK > orchestrate chat start
[INFO] - local tenant found
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Waiting for orchestrate server to be fully started and ready...
[INFO] - Starting docker-compose UI service inside LimalifecycleManager...
[INFO] - Chat UI Service started successfully.
[INFO] - Waiting for UI component to be initialized...
[INFO] - Opening chat interface at http://localhost:3000/chat-lite
[WARNING] - When using local chat, requests that the user 'Connect Apps' must be resolved by running `orchestrate conne
ctions set-credentials`
ADK > orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
[INFO] - Successfully imported knowledge base 'ibm_knowledge_base'
ADK > orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
```

- Wait for the message (A) indicating successful agent creation:

```
wxo-ADK - zsh - 119x31

Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
ADK >orchestrate agents import -f city-agent/city_agent.yaml
[INFO] - Agent 'city_agent' imported successfully
ADK >orchestrate chat start
[INFO] - local tenant found
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Waiting for orchestrate server to be fully started and ready...
[INFO] - Starting docker-compose UI service inside LimaLifecycleManager...
[INFO] - Chat UI Service started successfully.
[INFO] - Waiting for UI component to be initialized...
[INFO] - Opening chat interface at http://localhost:3000/chat-lite
[WARNING] - When using local chat, requests that the user 'Connect Apps' must be resolved by running 'orchestrate conne
ctions set-credentials'
ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
[INFO] - Successfully imported knowledge base 'ibm_knowledge_base'
ADK >orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
[INFO] - Agent 'ibm_agent' imported successfully (A)
ADK >
```

Agent testing

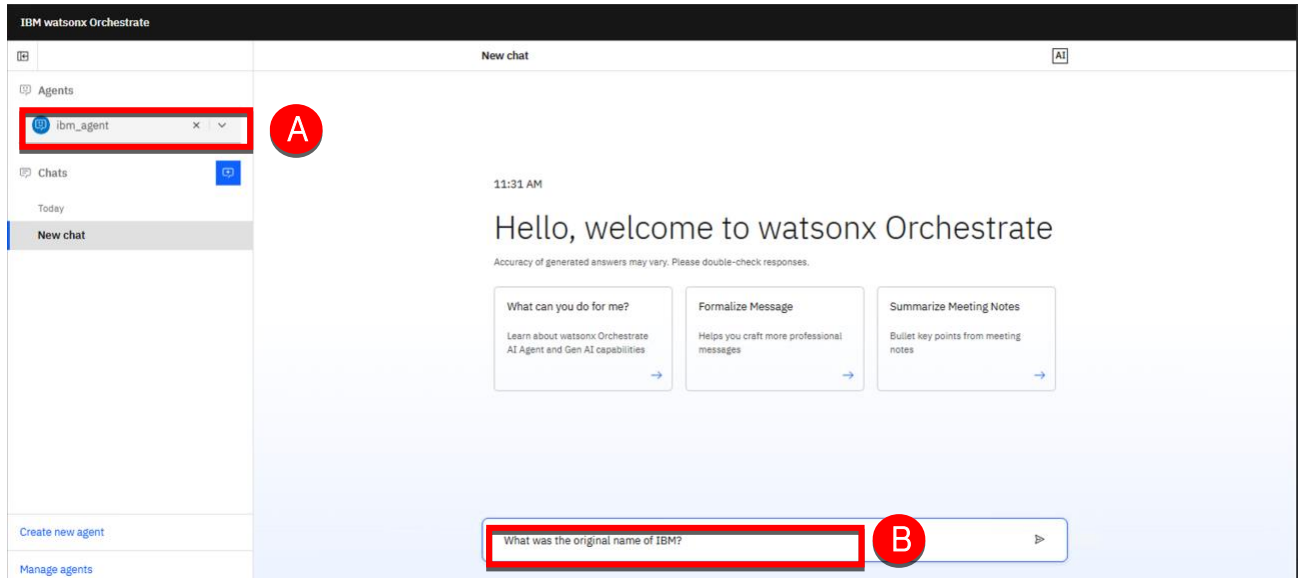
In this section you will start the chat interface, test the agent and review its reasoning in the chat UI.

- Enter the following command into the terminal (A) to start the chat server.

```
orchestrate chat start
```

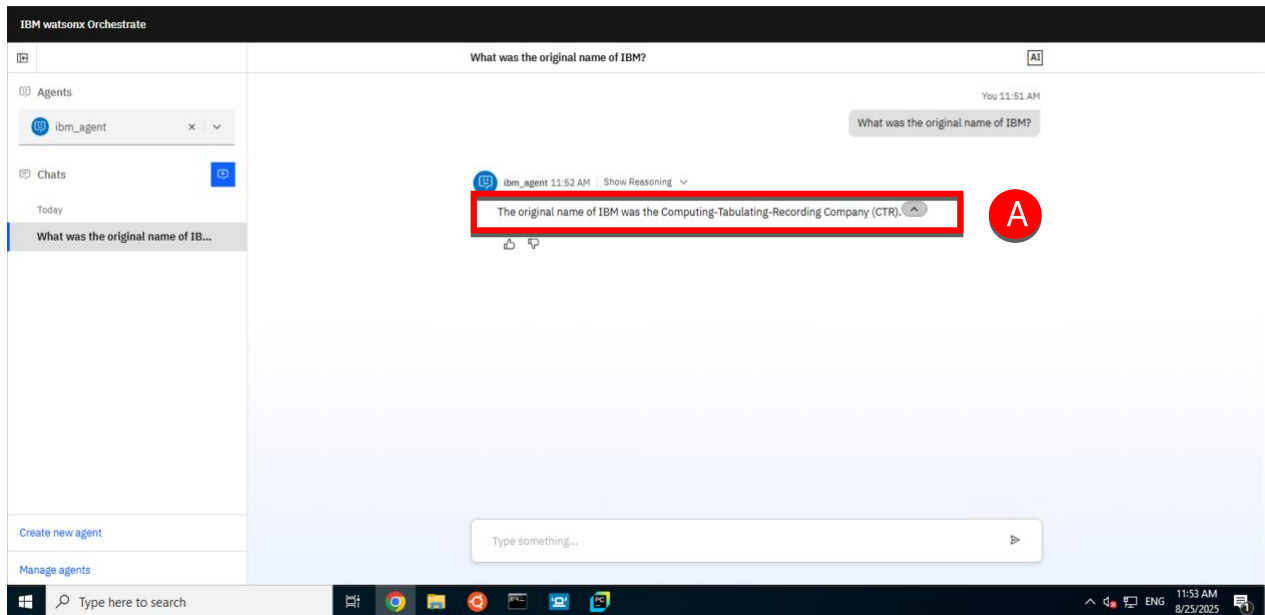
```
[INFO] - Agent 'city_agent' imported successfully
ADK >orchestrate chat start
[INFO] - local tenant found
[INFO] - Auto-detecting local IP address for async tool callbacks...
[INFO] - Auto-configured CALLBACK_HOST_URL to: http://9.83.134.220:4321
[INFO] - Waiting for orchestrate server to be fully started and ready...
[INFO] - Starting docker-compose UI service inside LimaLifecycleManager...
[INFO] - Chat UI Service started successfully.
[INFO] - Waiting for UI component to be initialized...
[INFO] - Opening chat interface at http://localhost:3000/chat-lite
[WARNING] - When using local chat, requests that the user 'Connect Apps' must be resolved by running 'orchestrate conne
ctions set-credentials'
ADK >orchestrate knowledge-bases import -f ibm_knowledge/knowledge_base/ibm_knowledge_base.yaml
[INFO] - Successfully imported knowledge base 'ibm_knowledge_base'
ADK >orchestrate agents import -f ibm_knowledge/agents/ibm_agent.yaml
[INFO] - Agent 'ibm_agent' imported successfully
ADK >orchestrate chat start (A)
```

- When the browser opens, ensure **ibm_agent** is selected (A) and enter "What was the original name of IBM?" into the chat window (B), then press Enter.



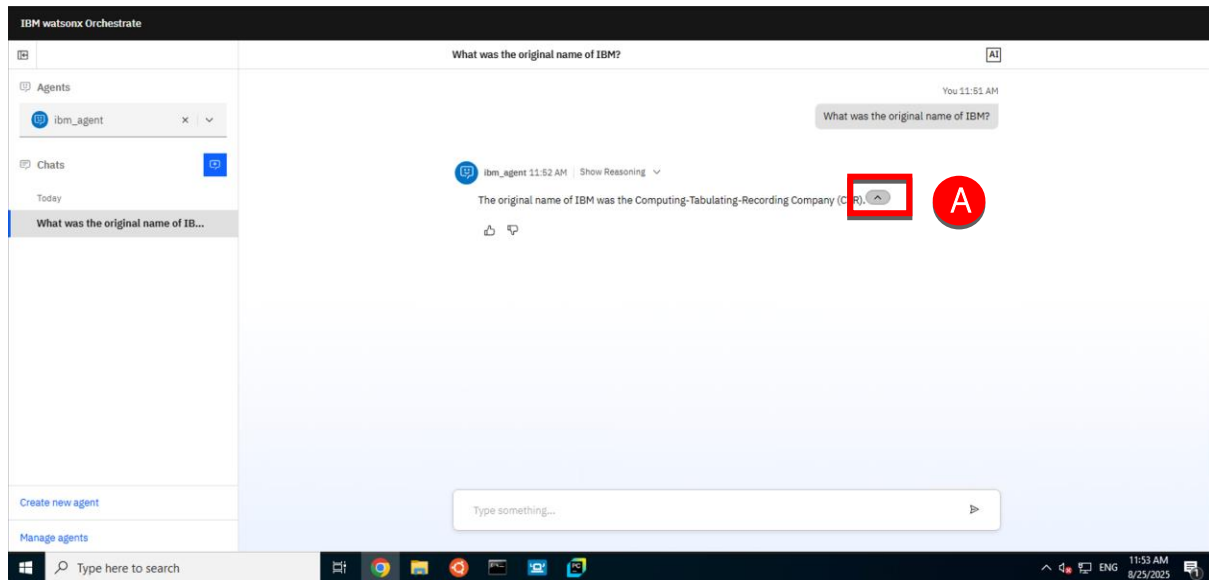
The user input is interpreted as a question about the IBM history. The agent will use the knowledge base that you created as the basis for the response. If the agent isn't visible, refresh your browser.

- After a few seconds the agent should reply (A):

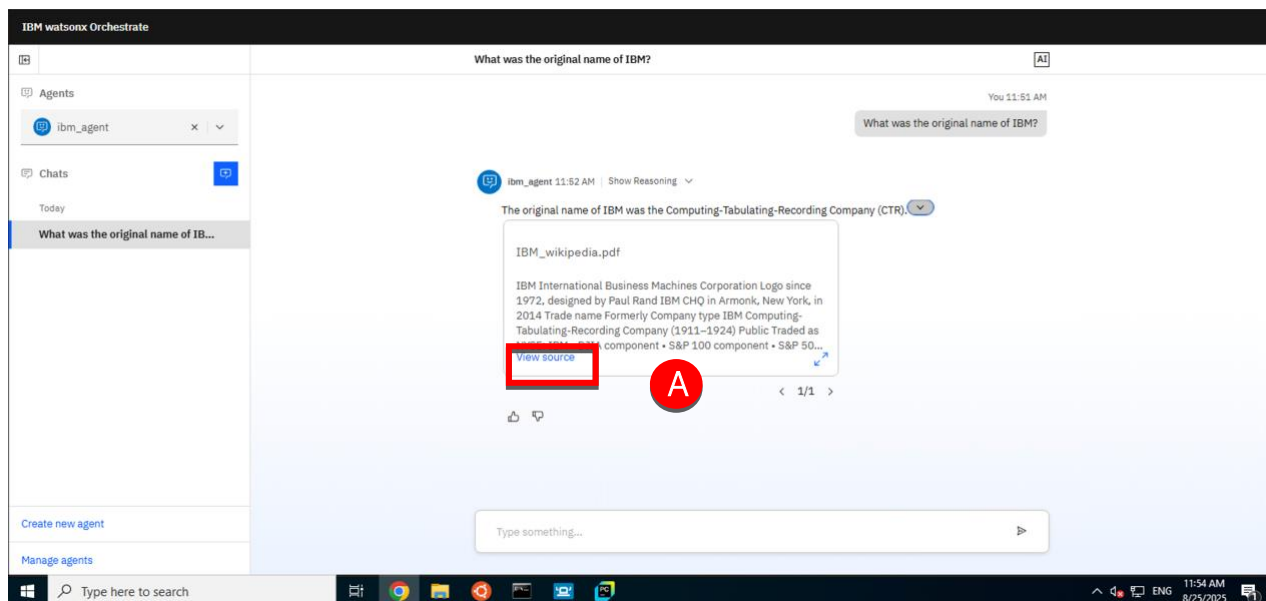


Please note that it can take 10-15 minutes for the citations to become available after uploading the documents.

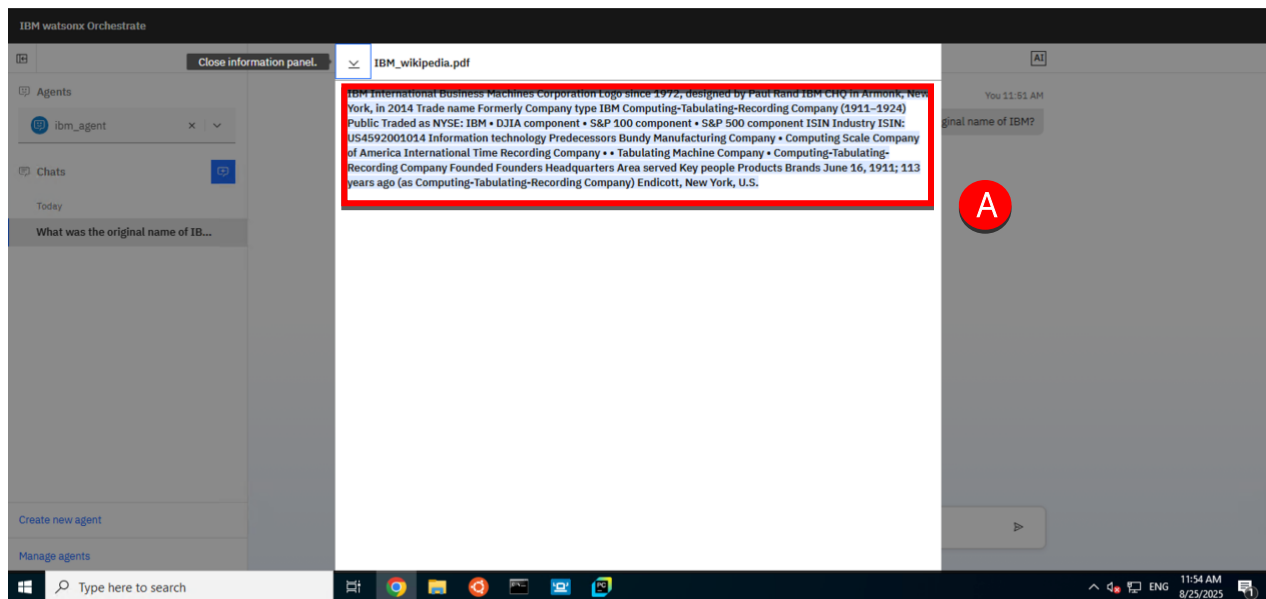
- Expand the citations panel (A).



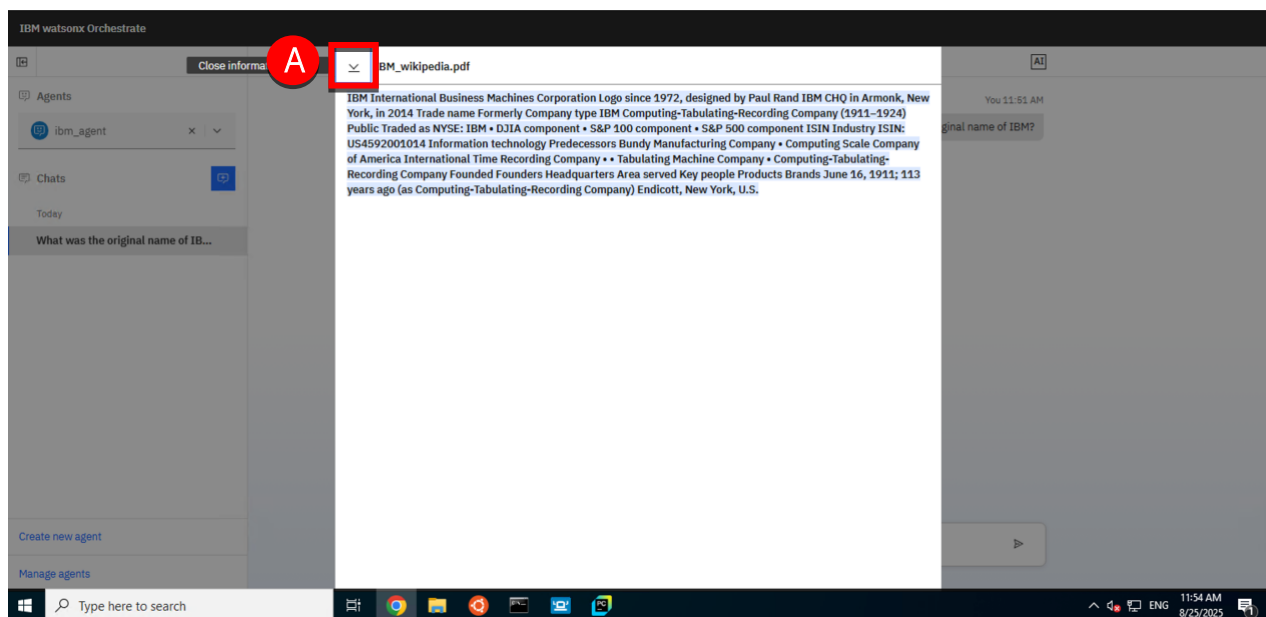
- Click View source (A).



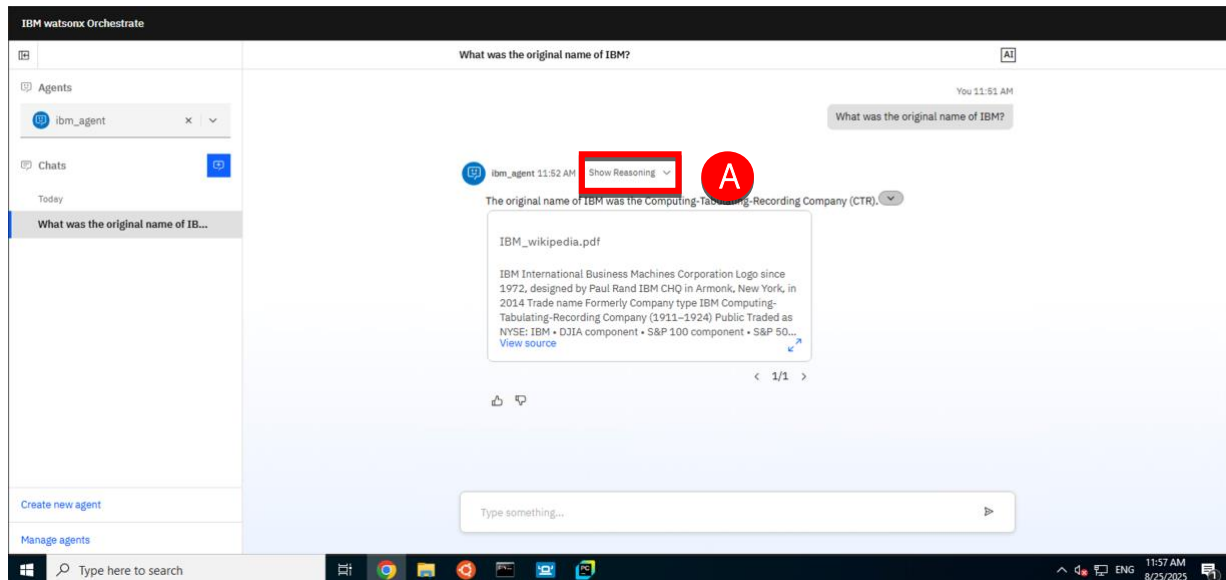
- Review the source of the answer from the document (A).



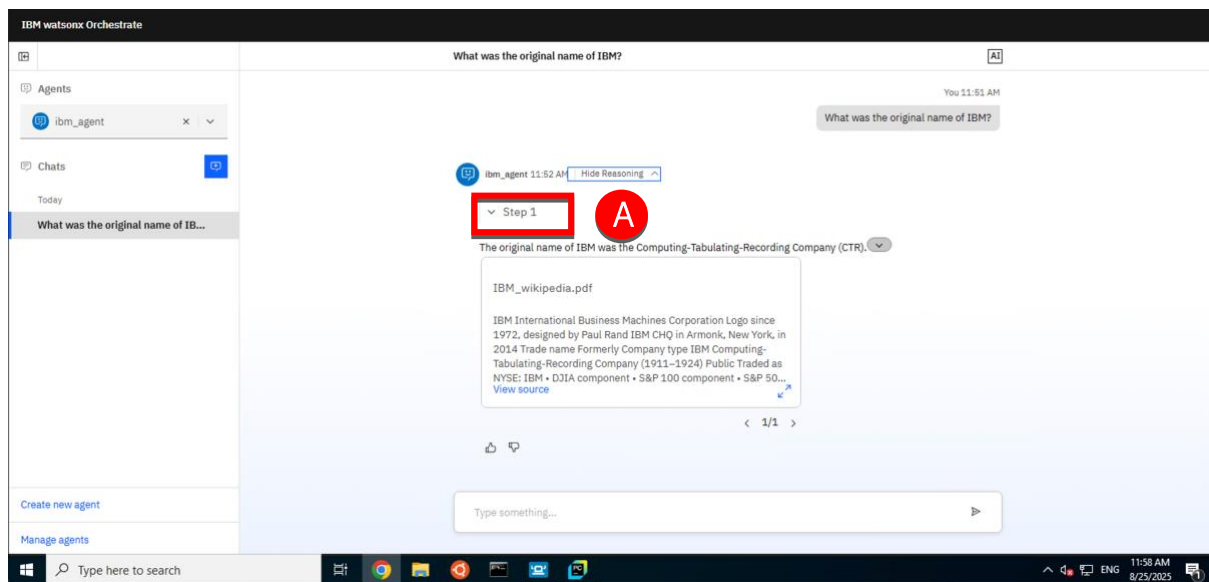
- Click the arrow icon to close the information panel (A).



8. Click Show Reasoning (A).



9. Expand the Step 1 panel (A).



10. The reasoning panel shows the input and output parameters (A) passed between the agent and the tool:

The screenshot displays the IBM watsonx Orchestrate interface. On the left sidebar, the 'Agents' section shows 'ibm_agent' selected. Below it, the 'Chats' section shows a chat titled 'What was the original name of IBM?'. The main area shows the reasoning panel for 'ibm_agent' at 05:22 PM. A red box highlights the 'Input - 1 Line' and 'Output - 1 Line' sections. The input is an empty JSON object `{}`. The output is the text: 'International Business Machines Corporation was originally founded as the Computing-Tabulating-Recording Company (CTR)'. A red circle with a white 'A' is placed next to the output text. Below the reasoning panel, the full text of the output is displayed: 'International Business Machines Corporation was originally founded as the Computing-Tabulating-Recording Company (CTR)'. At the bottom, there is a text input field with the placeholder 'Type something...' and a send button.

Summary

In this lab you created a knowledge base with topic specific documents and created an agent that was able to answer IBM history questions. The agent was able to adapt to the users request and interpret the information provided using the reasoning power of an LLM.

Appendix: Troubleshooting

System unresponsive

If your ADK environment becomes unresponsive use the command below to stop all running containers.

```
docker stop $(docker ps -q)
```

Enter the following command into the terminal to reset your server, then press Enter.

```
orchestrate server reset
```

Enter the following command into the terminal to restart your server, then press Enter.

```
orchestrate server start -l -e env
```

Enter the following command into the terminal (A) to start the chat server.

```
orchestrate chat start
```

You will need to repeat any commands to create tools and agents.

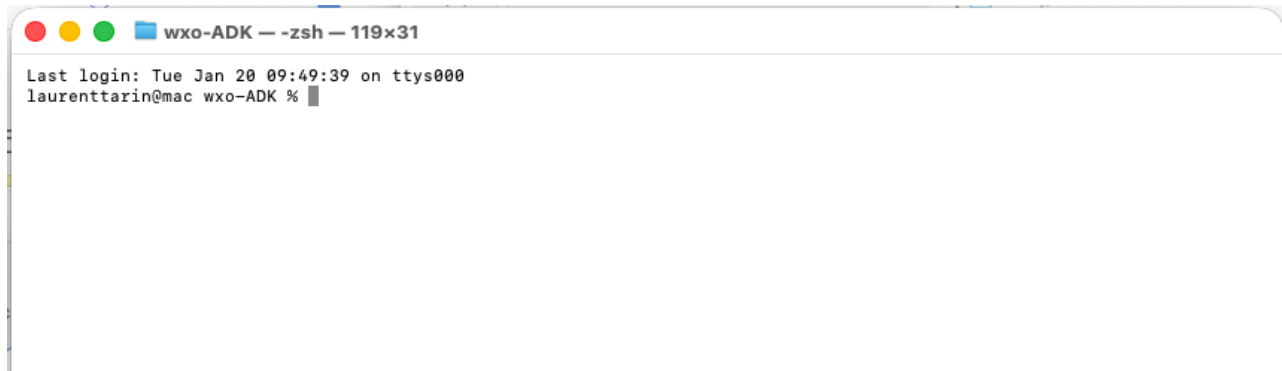
LLM Error

If you receive an error when using your agent (that worked previously), your API key may have been deleted. You will need to recreate the API key from your watsonx Orchestrate instance, update your env file, and restart the server.

Appendix: Reset your environment

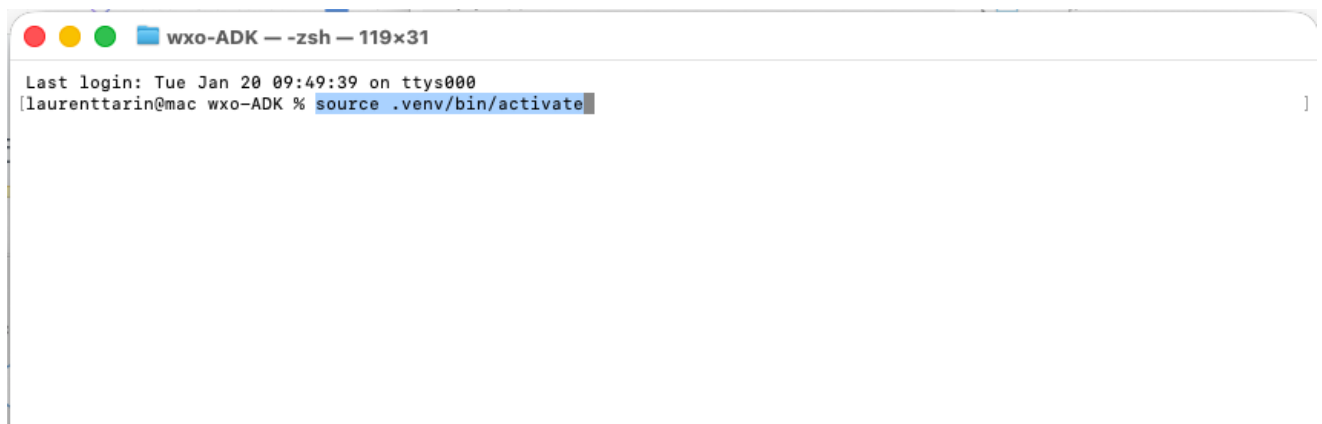
These steps will remove any existing tools and agents from your environment and restart the local watsonx Orchestrate server in readiness to start the lab.

1. Open a Terminal window on your **wxO-ADK** folder (A).



2. Enter the following command into the terminal to start your virtual environment:

```
source .venv/bin/activate
```



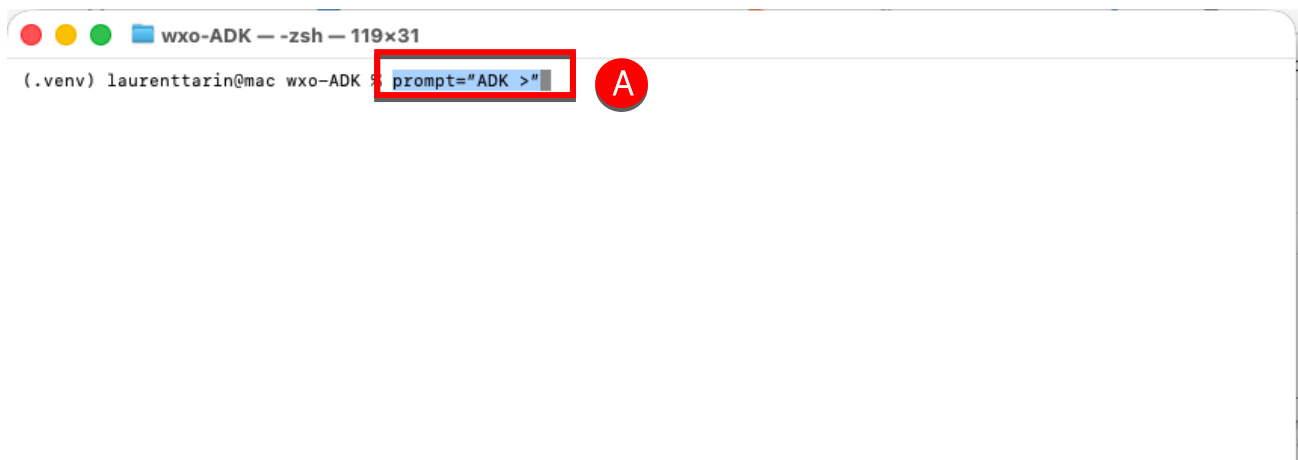
3. Check that the prompt starts with (.venv) (A) indicating that you are in your virtual environment.



```
wxo-ADK — zsh — 119x31
Last login: Tue Jan 20 09:49:39 on ttys000
[laurenttarin@mac wxo-ADK % source .venv/bin/activate
(.venv) laurenttarin@mac wxo-ADK %
```

4. Enter the following command into the terminal to improve the command visibility, then press Enter:

```
MS Windows users: function prompt{"ADK>"}
Mac users: prompt="ADK >"
```



```
wxo-ADK — zsh — 119x31
(.venv) laurenttarin@mac wxo-ADK % prompt="ADK >"
```

5. Enter the following command into the terminal to stop your chat service, then press Enter.

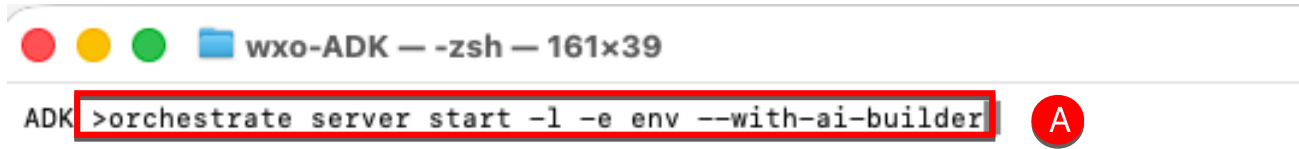
```
orchestrate chat stop
```

6. Enter the following command into the terminal to reset your server, then press Enter.

```
orchestrate server reset
```

7. Enter the following command into the terminal to start the server with your env configuration, then press Enter.

```
orchestrate server start -l -e env --with-ai-builder
```



Note: Some services may fail to start on the first attempt, and you may encounter issues connecting to the docker repository. Retry the start command in this situation, depending on the volume of requests, it can take several attempts to connect.

When startup is complete the terminal output will resemble the example below, restart can take 2-3 minutes to complete. The local server will be ready when you will see this **INFO** in the terminal screen (A):

```
wxo-ADK -- -zsh -- 119x31

Skipping already applied migration: 20251001.sql
Skipping already applied migration: 20251006.sql
Skipping already applied migration: 20251007.sql
Skipping already applied migration: 20251008.sql
Skipping already applied migration: 20251009.sql
Skipping already applied migration: 20251022.sql
Skipping already applied migration: 20251023.sql
Skipping already applied migration: 20251029.sql
Skipping already applied migration: 20251101.sql
Skipping already applied migration: 20251103.sql
Skipping already applied migration: 20251104.sql
Skipping already applied migration: 20251107.sql
Skipping already applied migration: 20251110.sql
Skipping already applied migration: 20251111.sql
Skipping already applied migration: 20251112.sql
Existing wxo_observability DB found
Skipping already applied observability migration: 00_init_db.sql
Skipping already applied observability migration: 01_create_tables.sql
Skipping already applied observability migration: 02_triggers_and_functions.sql
Skipping already applied observability migration: 03_security.sql
Skipping already applied observability migration: 04_rls_policies.sql
Skipping already applied observability migration: 20250116.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - local tenant found
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the
UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3010, username: orchestrate@ibm.com, pa
ssword: orchestrate
```



You are now ready to work with watsonx Orchestrate ADK.

